Team no. : 44

Team name : Logic Loop

D5 – Fintech transaction Anomaly & fraud Visualizer

Insight Report (D5 Hackathon Project)

1. Introduction & Problem Statement

In today's digital financial ecosystem, fraud detection has become one of the most critical challenges. With the rapid increase in online transactions, identifying fraudulent activities in real-time is difficult due to large volumes of data, hidden patterns, and evolving fraud techniques.

The dataset provided for this hackathon represents financial transaction logs containing details such as user activity, timestamps, transaction amounts, risk scores, and locations.

Core Problems Identified:

- Difficulty in identifying **real-time fraudulent transactions**
- Lack of visibility into **high-risk users and locations**
- No clear way to detect **unusual transaction behavior (velocity, frequency)**
- Inefficient monitoring of **fraud patterns over time**
- Missing insights for **decision-making and prevention strategies**

2. Objectives of the Project

The main goal of our project was to build an **interactive and intelligent fraud detection dashboard** that:

- Detects suspicious activities in real-time
- Highlights high-risk users and locations
- Tracks transaction behavior patterns
- Provides actionable insights for fraud prevention
- Improves decision-making using data visualization

3. Approach & Methodology

To solve the problem effectively, we followed a structured approach:

Step 1: Data Understanding

We analyzed the dataset to understand:

- Transaction flow
- User behavior patterns
- Risk scoring system
- Time-based activity trends

Step 2: Feature Engineering (Key Columns Created)

We created several calculated columns to improve fraud detection accuracy:

1. Transaction_Count_5Min

- Counts number of transactions by a user in 5 minutes
- Helps detect **rapid activity (bot/fraud burst)**

2. Transaction_Count_1Hr

- Tracks hourly activity
- Identifies **abnormal transaction spikes**

3. Velocity_Flag

- Flags users with unusually high transaction frequency
- Detects **suspicious speed of transactions**

4. Risk Level Classification

- Categorized transactions into:
    - Low Risk
    - Medium Risk
    - High Risk

5. Time-based Features

- Extracted hour, day patterns
- Helps analyze **fraud timing trends**

Step 3: Dashboard Design

We designed a clean and interactive dashboard using charts, KPIs, and filters.

Key Visual Components:

- KPI Cards (Total Transactions, Fraud Rate, High Risk Count)
- Line Charts (Transactions over Time)

- Bar Charts (Risk Level Distribution)
- Heatmaps (City-based risk analysis)
- Tables (Detailed transaction logs)

4. Problems Solved

✅ 1. Real-Time Fraud Detection

Problem: No real-time monitoring
Solution: Implemented live transaction tracking and updating metrics
Impact: Immediate detection of suspicious activities

✅ 2. Hidden Fraud Patterns

Problem: Fraud patterns were not visible
Solution: Used time-series and frequency analysis
Impact: Identified peak fraud hours and unusual spikes

✅ 3. High-Risk User Identification

Problem: Difficult to detect risky users
Solution: Created velocity and transaction count indicators
Impact: Quickly highlights suspicious users

✅ 4. Location-Based Fraud

Problem: No geographical insights
Solution: Built city-wise risk visualization
Impact: Identified high-risk cities

✅ 5. Data Interpretation Complexity

Problem: Raw data was hard to understand
Solution: Converted into interactive visual dashboard
Impact: Easy decision-making for stakeholders

5. Key Insights Discovered

From our analysis, we discovered several important insights:

📊 1. High Transaction Velocity = High Fraud Risk

Users performing many transactions in a short time are more likely to be fraudulent.

📊 2. Specific Cities Show Higher Risk

Certain locations consistently show higher fraud activity, indicating possible fraud hubs.

📊 3. Peak Fraud Timing

Fraud transactions tend to occur during specific hours (late night or unusual times).

📊 4. Repeated Patterns

Fraudsters often repeat patterns such as:

- Same transaction amount
- Rapid consecutive transactions
- Similar timestamps

📊 5. Risk Score Correlation

Higher risk scores strongly correlate with:

- High transaction frequency
- Abnormal behavior patterns

6. Technical Implementation

Frontend

- HTML, CSS, JavaScript
- Interactive charts and UI
- Real-time updates

Backend

- Python (Flask)
- Data processing and API handling
- Real-time data simulation

Tools Used

- Power BI (for dashboard logic)

- Python (data handling)
- CSV-based real-time simulation

7. What We Have Achieved

- Built a **complete fraud detection system**
- Created **real-time monitoring dashboard**
- Identified **high-risk users and locations**
- Developed **smart indicators for fraud detection**
- Delivered a **user-friendly and interactive solution**

7. Fraud Detection and Risk Analysis Dashboard

Page 1: Executive Overview (Command Center)

What We Built

- A complete **Fraud Detection Dashboard**
- Shows revenue, risk, transactions in one place
- Helps monitor fraud in real-time

Key Metrics (From Dashboard)

- Total Revenue: **~2.85M**
- Total Transactions: **~33K**
- Success Rate: **94% Approved**
- Fraud/Failure Rate: **Very Low (0%)**
- Avg Fraud Risk Score: **~1.06**

Insights

- Most transactions are safe (94% approved)
- Fraud rate is low but still needs monitoring
- Revenue is high → system is active and large-scale

Problem Solved

- No central view → Now we have **one command center**
- Hard to track metrics → Now easy with KPIs

Page 2: Fraud Intelligence Analysis

What This Page Shows

- Fraud rate trends
- Failed transactions over time
- Risk score by city

Key Findings

- Failed transactions: **669 (Year 2023)**
- Fraud rate is **very low but consistent**
- Cities like **Sydney, London, Mumbai** show higher risk

Insights

- Fraud is not random → it follows patterns
- Certain locations have higher fraud probability
- Fraud rate changes daily (time-based behavior)

Problem Solved

- Earlier: Fraud patterns hidden
- Now: **Clear fraud trends visible**

Page 3: User Behavior Analysis

What This Page Shows

- User activity
- Spending behavior
- Transactions per user

Key Metrics

- Total Users: **13K**
- Avg Spend/User: **~208.97**
- Transactions/User: **~2.5**
- High Risk Transactions: **Very Low**

Insights

- Most users behave normally

- Few users generate high transactions → possible fraud
- Some users dominate revenue (top contributors)

Problem Solved

- Earlier: No user tracking
- Now: **Identify suspicious users easily**

Page 4: Geo Intelligence (Location Analysis)

What This Page Shows

- Fraud by city
- Revenue by city
- Risk comparison

Key Findings

- Total Cities: **6**
- Top Risk City: **Sydney**
- Transactions evenly distributed (~33K each)

Insights

- Fraud is location-dependent
- Some cities consistently show higher risk
- Revenue is similar across cities but risk differs

Problem Solved

- Earlier: No location insight
- Now: **Find fraud hotspots**

Page 5: Category Analysis

What This Page Shows

- Performance by category (Groceries, Retail, Crypto, etc.)

Key Metrics

- Total Revenue: **17.04M**
- Total Categories: **6**
- Top Category: **Groceries**

Insights

- Groceries and Retail have highest transactions
- Crypto & Electronics show higher risk levels
- Some categories have higher failure rates

Problem Solved

- Earlier: No category-level fraud detection
- Now: **Know which business area is risky**

Page 6: Real-Time Monitoring

What This Page Shows

- Live transactions
- Live fraud rate
- Alerts and activity

Key Metrics

- Live Failed Transactions: **669**
- Active Users: **15K**
- Daily Transactions: **~293K**

Insights

- System supports **real-time tracking**
- Fraud can be detected instantly
- Live alerts help quick action

Problem Solved

- Earlier: Static data only
- Now: **Real-time fraud detection system**

Page 7: Fraud Command Center (Advanced Insights)

What This Page Shows

- Fraud patterns
- Velocity detection (fast transactions)
- Risk comparison

Key Metrics

- Total Transactions: **200K**
- Fraud Count: **669**
- Active Users: **15K**

Key Insights

- High transaction speed = High fraud chance
- Fraud happens in spikes (not constant)
- Cities have similar risk but slight variation

Advanced Findings

- Velocity detection helps catch bots
- Fraud timeline shows peak activity periods
- Category vs failure shows risk relationships

Problem Solved

- Earlier: No deep analysis
- Now: **Advanced fraud detection + prediction**

9. Conclusion

This project successfully transforms raw financial transaction data into a powerful fraud detection system. By combining data analysis, visualization, and real-time processing, we solved key challenges in fraud detection.

Our dashboard not only identifies fraud but also provides meaningful insights that can help organizations **prevent financial losses, improve security, and make better decisions**.

10. Future Improvements

- Integration with **machine learning models**
- Automated fraud alerts (notifications)
- Advanced anomaly detection
- Integration with live banking systems
- Predictive fraud analysis